

T16 Pre-Flight and Post-Flight Check & Flight Test

I. Pre-Flight Checklists

1. Structural Inspection

- Unfold the aircraft arms to check whether the thread sleeve is mounted too tight. Manually rotate the motor base to the left or right to check if the base is loose.

2. Motor Check

- Manually rotate the motor to check whether it can rotate smoothly. Check whether the propellers are broken, and check whether the CCW propellers are attached to the motors 1, 3, and 6 while CW propellers are to the motors 2, 4, and 5.

3. Battery Check

- Check whether there is debris on the battery socket (gold), check if the socket is misshapen, and check if the battery is connected well.
- Check whether the battery is mounted firmly and make sure that the battery does not come off during the flight.

4. Antenna Check

- Check whether the RTK antenna and SDR antenna are attached firmly and not broken.
- Check whether the antenna connector is attached to the aerial-electronics system board firmly.

5. Radar Check

- Check whether the radar is blocked by foreign objects, check whether the radar's external shell is broken, and check whether the radar bracket is not mounted firmly enough or damaged.
- Ensure the radar's installation direction is to the left.

6. Motor Angle Check

- In the T16's original settings, the motor base's position has already been adjusted so that the motor tilt angle is 5° horizontally and 4° vertically. The user can directly mount the motor to the motor base. Ensure that the arms 4 and 1 are on the same level, arms 2 and 6 are on the same level, and arms 3 and 5 are on the same level.

- Check whether the pitch angle of the M1 motor is the same as that of the M4 motor, the pitch angle of the M2 motor is the same as that of the M6 motor, and pitch angle of the M3 motor is the same as that of the M5 motor. (The direction shall be the direction of the aircraft nose.)

No. 1 motor tilts horizontally to the left by 0°.

No. 2 motor tilts backward horizontally by 5°.

No. 3 motor tilts forward horizontally by 5°.

No. 4 motor tilts horizontally to the left by 0°.

No. 5 motor tilts forward horizontally by 5°.

No. 6 motor tilts backward horizontally by 5°.

7. Propulsion System Check

- Perform the motor idle speed test by using DJI Assistant 2 and check whether the motor rotates in the correct direction and does not make abnormal sounds. Makes sure that the motors 1, 3, 6 rotate counterclockwise and motors 2, 4, 5 rotate clockwise.
- Check whether the propellers are mounted correctly. Ensure that the CCW propellers are attached to the motors 1, 3, and 6 while CW propellers are to the motors 2, 4, and 5.

8. Firmware Version Check

- Check that the aircraft's firmware version, remote controller's firmware version, and app version are all the most recent versions.

9. Dongle Check

- Insert the 4G wireless dongle into the remote controller and wait for about one minute. Then go to Network Diagnostics in the app to check whether the "DJI Agriculture Management Platform" option is displayed as "Normal".

10. Remote Controller Built-in GPS Satellite Search

- Launch the app, go to "Field Plan", and then tap "Walk with RC". Wait for two minutes, and ensure that the satellite count is more than 10 and that the GPS positioning accuracy is within 2 m. Pay attention to the GPS satellite icon displayed on the app during the operation.

11. Remote Controller Control Stick Calibration

- Go to RC Settings and Control Stick Calibration. Then calibrate the control sticks following the app instructions. When pushing the control sticks, observe the amount of the rod displayed on the app.

12. Linking

- Launch the app, go to RTK Settings, and then tap “Linking”. After completing the D-RTK 2 mobile station linking, click on the bullet box to start linking the remote controller with the aircraft.
After linking, if the remote controller is used again to link to a different aircraft, a disconnection warning will be shown on the RTK Settings interface. Click the settings interface to relink.

13. LED Check

- Power on the aircraft and check whether the M3/M4/M5 ESC LEDs are solid red, green, and red respectively, and also check whether the brightness is at a normal level.

14. Sensor Calibration

- Compass Calibration, Method 1: Go to “Aircraft Settings”, “Advanced Settings”, “Sensors”, and tap “Calibrate Compass”. (Calibration is not required if there is no app prompt.)
- Compass Calibration, Method 2: Enter the compass calibration interface by turning the remote controller F/A/P switch back and forth 3 times or more.

15. Battery Information Check

- Launch the app and go to “Aircraft Battery”. Observe the cell voltage, total voltage, temperature, current, serial number display, and then enable “Show Voltage on Main Screen”.

16. Spray Control Check

- Insert the fully loaded spray tank into the aircraft. Ensure that the spray tank and connection between the spray tank’s one-way valve and the aircraft do not leak.
- After the spray tank is filled with water and inserted into the aircraft, the “Remaining Pesticide” under Spraying System Settings in the app should show “>8.5L”. If the tank has half of its capacity, “Remaining Pesticide” will be shown as “>1.3L” or “1.6L”.
- Press the Spray button on the remote controller to start spraying, and then the pump can spray water. Press the button again to stop spraying, and the pump will stop working.

- After the spraying has started, the nozzle can spray the water efficiently without any forking or blockage.
- Ensure that the angle of the water outlet direction of the eight nozzles is 90° with the aircraft's landing gears.
- Unscrew the pressure relief valve on the spray lance to discharge all the air in the pipelines.

“Spray System Detection” in “Spray System Settings” can independently control pump operations. For example, when sliding the No. 1 delivery pump speed slider, then the No.1 delivery pump operates.

- The Spray Rate dial on the left side of the remote controller can adjust the flow rate. Press the Spray button, adjust the dial to the left, and then the flow will become smaller; continually adjusting the dial to the left reduces the number of spraying nozzles to only four nozzles. Adjust the dial to the right, and the flow will become larger, and continually adjusting the dial to the right will make eight nozzles start spraying.
- After pressing the Spray button on the remote controller, the corresponding OSD data will be shown on the app (flow rate: L/min).
- Calibrate the delivery pump.

17. Drone Satellite Search

- Place the aircraft in an open, unobstructed area. Satellite counts should be no less than 10. Then the drone will enter GPS mode and automatically refresh the Home Point (M3/M4/M5 LED blinks green quickly). Enable the RTK function, and check whether there is a RTK FIX prompt shown on the app. Go to RTK Settings and check whether Positioning is shown as “NARROW-INT” and Orientation is shown as “FIX”. Then check the dual antenna GPS, BeiDou, GLONASS, and Galileo data.

II. Flight Tests

Tips: Before taking off for high-altitude intensive flight operations, please turn off “Set Height” and “Auto Obstacle Avoidance”, and also turn on “Obstacle Avoidance Radar”.

1. Turn on “Front lamp brightness” in the app to see if the light is bright and that there is no significant difference between the light brightness on both sides. During the flight, check whether the FPV screen freezes or is snowy.
2. Set the control stick mode and push the sticks to make the drone take off.
3. Push the pitch stick up or down to make the aircraft fly forwards or backwards. Push the throttle stick up or down to make the drone ascend or descend. Push the roll stick left or right to let the drone fly left or right. Check whether the drone can fly following the commands and whether the drone flies in a straight line.
4. When the drone hovers in place, push the yaw stick to rotate the drone counterclockwise or clockwise. Ensure that the drone does not drift more than 0.5 m and that the drone does not shake during rotation.
5. When “Obstacle Avoidance Radar” is enabled, fly the drone (speed < 4 m/s) to the front or back of an obstacle to check whether the radar obstacle avoidance is functional.
6. Check the flight spraying control function: Fill the spray tank with water, take off the drone, press the Spray button to start spraying, and toggle the Spray Rate Dial to decrease the flow rate to the minimum. When the aircraft’s nose faces forward, the four nozzles at the tail of the drone will start spraying. When the tail faces forwards, the four nozzles at the aircraft’s nose will start spraying.
7. Disable the RTK function and fly the drone to 8 m above the ground. Then make the drone hover in place for 30 seconds and check whether the drone’s flight height drops. During the hovering, check whether the drone or landing gear shakes. Ensure that the height does not fluctuate more than ± 0.5 m and that the drone does not drift more than half the distance of its frame (for example, if the drone’s frame is 2.5 m long, it should not drift left or right more than 1.25 m).
8. Disable the RTK function and make the drone fly forward, backward, to the left, and to the right for a distance of 30 m each, and check whether the drone’s flight height drops. The drone’s altitude should not fluctuate more than ± 0.5 m. Ensure that the drone can brake normally and that the drone does not shake during flight.
9. In Point A/B mode, ensure that the “Set Height” function is disabled, the water volume of the spray tank is more than 5 L, the route length is longer than 30 m, the line spacing is 3 m, the flight speed is no less than 5 m/s, and that the number of flight routes is 3. Then check whether the drone can fly stably when accelerating or decelerating.
10. Enable “Set Height” in the app, enter Manual Plus (M+) mode, set the flight height to 2.5 m, and check the flight for 20 seconds to see if the drone altitude fluctuates more than ± 20 cm. Push the throttle stick forward to ascend the drone to 4 m, and release the stick to see if the drone can descend back down to 2.5 m. Then hover the drone for 10 seconds and turn off “Set Height” after the test.

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